

Shazil Shiraj Katchhi

AI & ML Engineer | Computer Vision | LLM Applications

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Berlin, Germany



PROFESSIONAL SUMMARY

M.Sc. Artificial Intelligence student at BTU Cottbus-Senftenberg with deep hands-on experience in Machine Learning, Deep Learning, Computer Vision and LLM-based application development. IEEE-published researcher - real-time sign language translation system achieving 91% gesture accuracy. Smart India Hackathon Finalist (Top 5). Skilled in PyTorch, TensorFlow, OpenCV, Python and LLM API integration. Eligible to work as a Werkstudent in Germany - up to 20 hours/week during semester.

EDUCATION

M.Sc. Artificial Intelligence Mar 2026 – Present

Brandenburg University of Technology (BTU) Cottbus-Senftenberg, Germany

Coursework: Machine Learning · Deep Learning · Computer Vision · Data Analytics · Statistical Methods

B.E. Computer Science & Engineering (AI & ML) 2021 – 2025

M.H. Saboo Siddik College of Engineering, University of Mumbai, India

CGPA: 7.14 / 10 | Coursework: Machine Learning · Computer Vision · AI · Data Structures · DBMS

Higher Secondary Certificate (HSC) 2019 – 2021

Navneet College of Commerce, Arts & Science, Mumbai, India

Score: 89.6%

PROFESSIONAL EXPERIENCE

Data Analyst Intern Jun 2024 – Oct 2024

Infosys Limited, Bangalore, India

- Built ETL pipelines using Python, SQL and cloud services (AWS/Azure); automated data processing and reporting workflows to reduce manual effort and improve reproducibility.
- Designed Power BI and Tableau dashboards for real-time KPI monitoring; optimised SQL queries via indexing and partitioning.

Raspberry Pi Engineer Intern Feb 2024

MHSSCE Alumni Startup, Mumbai, India

- Built an IoT sensor data pipeline using Raspberry Pi and Node-RED - real-time fuel density and GPS data from field sensors to a live dashboard, fully automated.

RESEARCH PUBLICATION

Empowering Deaf Communication: A Deep Learning Framework for Mar 2025

Sign-to-Speech Translation

IEEE Xplore · ESCI 2025 · AISSMS IoT, Pune · Co-Author

- Co-designed a real-time ISL-to-speech/text system combining Transformer, CNN (ResNet) and LSTM with MediaPipe - 91% gesture-detection accuracy.
- Deployed on Raspberry Pi with inference latency optimisation; integrated multilingual NLP and TTS; evaluated with systematic ground-truth test scenarios.

PROJECTS

Smart Glasses for Speech & Hearing Impaired - Smart India Hackathon Dec 2024 – Mar 2025

Finalist (Top 5) [GitHub](#)

Tech: Python · MediaPipe · CNN (ResNet) · Vision Transformer (ViT) · LSTM · Raspberry Pi · TTS

- End-to-end real-time AI pipeline - camera > MediaPipe pose extraction > ViT+CNN classification > LSTM temporal sequencing > TTS output. Deployed on Raspberry Pi with latency optimisation.
- Systematic evaluation: ground-truth labelling, per-class detection benchmarking across gesture classes.

Credit Card Fraud Detection [GitHub](#)2024

Tech: Python · Scikit-learn · XGBoost · Pandas · NumPy

- Built an ML classification pipeline (Logistic Regression, Random Forest, XGBoost) on an imbalanced dataset; applied SMOTE and evaluated using F1-score and AUC-ROC.

TEXTURA - Prompt-Based Image Generation [GitHub](#)2025

Tech: Python · Diffusion Models · LLM APIs (Claude, GPT)

- Engineered a text-to-image pipeline via LLM API integration with structured prompt engineering, model evaluation and a user-facing interface.

Object Detection System [GitHub](#)2024

Tech: Python · TensorFlow · OpenCV

- Developed and benchmarked a real-time object detection model; optimised precision/recall trade-offs and inference speed across hardware configurations.

Open Chatroom Web Application2024

Tech: React.js · Node.js · Firebase

- Built a full-stack real-time communication app with Firebase auth, cloud data sync and collaborative note sharing.

TECHNICAL SKILLS	
Programming	Python, SQL, C++, R, JavaScript
AI / ML	Machine Learning, Deep Learning, Computer Vision, NLP, Object Detection, CNNs, Transformers (ViT), LSTMs, Diffusion Models, Fine-Tuning, Generative AI
LLM / GenAI	Claude API, GPT API, Prompt Engineering, LLM Integration, Rapid Prototyping, GitHub Copilot
Libraries	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, OpenCV, MediaPipe, Pandas, NumPy, Matplotlib, Seaborn
Data & Cloud	ETL Pipelines, SQL Optimisation, Power BI, Tableau, AWS, Azure, Docker
Databases	MySQL, PostgreSQL, SQLite, SQL Server
Web & IoT	React.js, Node.js, Django, Firebase, Node-RED, Raspberry Pi, Embedded ML
Tools	Git, Jupyter Notebook, VS Code, Linux, Postman, Figma

LEADERSHIP & ACTIVITIES	
• Chairperson - IEEE MHSSCE Student Chapter	2024 – 2025
• General Secretary - SACE Committee, MHSSCE	Jan – Jun 2025
• Social Media Head - IEEE MHSSCE Student Chapter	Mar – Dec 2023
• Graphics Team Lead - SACE Committee, MHSSCE	Feb – Apr 2023

LANGUAGES	
English	C2 – Proficient (IELTS 8.5)
German	A2 – Elementary (working towards B1)
Hindi	Native
Urdu	Native

ADDITIONAL INFORMATION	
Work Auth	Eligible to work as a Werkstudent in Germany
Availability	Up to 20 hrs/week during semester · Full-time during breaks
Interests	AI, Machine Learning, Computer Vision, LLM Applications, Embedded AI, NLP, Autonomous Systems